ENGR 390
Information and Communication Technology for Sustainability (ICT4S)
Syllabus
Spring 2019

Professor: Dr. Birgit Penzenstadler
Class meetings: Mon/Wed lecture 9:00-9:50, activity 10:00-10:50
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Office Hours: Mon 12-1pm and per individual appointment (easiest per email)

DESCRIPTION

Catalogue description: Concepts for understanding and analyzing the sustainability of a given engineering context and how to support it by information and communication technology. Application domains relevant for sustainability engineering like climate change, smart systems, and gamification. Systems thinking for future scenarios.

In a few more words: This course aims to equip students with the foundational knowledge on sustainability and its relation to information and communication technology (ICT). It gives insights into the most important concepts for understanding, analyzing, and assessing the sustainability of a given engineering context. As engineers we are responsible for the long-term effects and impacts of the systems we develop; therefore, it is crucial that you develop an understanding of how these systems impact the environment and our society locally and globally.
Have you ever wondered about how climate change, energy production, and food consumption are related? Are you curious how you can analyze the big picture of sustainability with a few simple tools instead of having to rely on hearsay? How you can crosscheck statements about sustainability and whether we need to save our planet?
We will go through a list of application domains that are highly relevant for sustainability engineering, including but not limited to climate change, sustainable food production, smart systems, and gamification. Furthermore, we will learn about and experiment with systems thinking approaches and learn how to develop feasible future scenarios. This course is a general education writing-intensive capstone course. Expect a lot of active discussion in class as I welcome all your questions – I strongly believe we learn better in dialogue and interaction.

Lecture: 2 hours per week.
Activity: 2 hours per week.
Grading: Individual and team assignments. Letter grade only (A-F).

COURSE OBJECTIVES

- For "**Sustainability**":
    - **Understanding of basic sustainability concepts and application domains:** You develop the skill to explain the coarse history of and current application domains for sustainability concepts and can transfer them to a new context, e.g. in a case study.
    - **Sustainability dimensions in technology:** You develop the ability to critically reflect upon the different dimensions of sustainability in a given context, specifically in any context that involves software-intensive systems or information and communication technology.

- **Sustainability assessment:** You develop the skills to analyze a given technological solution for its benefits and drawbacks with regard to the sustainability of the surrounding application context and propose improvement measures.

- For "**Written Communication**": You enhance your writing skills with regard to the content development, disciplinary conventions, organization, language, global self-awareness, perspective taking, and understanding global systems – all in the context of sustainable development and ICT for sustainability.

- For "**Oral communication**": The ability to critically reflect upon the different dimensions of sustainability in a given context, specifically in any context that involves software-intensive systems or information and communication technology.

- "**Critical Thinking**": Ability to define sustainability in a specific context with regard to what to sustain, for whom, for how long, and at what cost. You will be able to define system scopes and understand causes of and dependencies between sustainability problems by seeing the bigger picture, by being able to expand a system's scope to include further impact factors.

- "**Inquiry and Analysis**": Ability to judge the quality of ICT solutions because you understand the wider impact these solutions may have on their operational environment and business as well as societal context. You can analyze the availability or scarcity of resources and judge their consequences for ICT development and its wider context in business, society, and for the environment.

- "**Social Responsibility and Civic Engagement**": Ability to translate sustainability concepts for their community and become visionaries and thought leaders that inspire actions. You can design sustainable solutions for their communities that fit the actual underlying needs.

COURSE TOPIC OUTLINE

This is a brief topic outline, but there is a more detailed table below with a timeline.

1. Introduction to sustainability (definitions, history, aspects)
2. Information and communication technology in relation to sustainability: Reduce-reuse-recycle, e-waste, planned obsolescence
3. The five dimensions of sustainability: environmental, individual, social, economic, and technical
4. The three orders of effect of technical systems and the rebound effects they can create by their operation
5. Peak oil and climate change and how they are facilitated and counteracted by software-intensive systems
6. Sustainable food production and permaculture facilitated by sensors, actuators, and database systems
7. Smart X – green product and service development using embedded systems and the internet of things
8. Behavior change: gamification of sustainability by developing game apps and web games for educating
9. Sustainable development concepts in engineering: sustainability design principles for engineers
10. Sustainable development process: human-centered, eco-aware systems design for long-term resilience
11. Introduction to systems thinking: computational thinking versus systems thinking and when to use which
12. System dynamics: understanding the stocks, flows, buffers, and feedback loops in systems

13. Leverage points: analyzing system dynamics and using leverage points to change systems sustainably
14. Sustainability assessment (metrics and procedures) for information and communication technology systems
15. Possible future scenarios for ICT systems that support different dimensions of sustainability

FURTHER READING

All resources that you need for this course will be posted on BeachBoard. You don't have to buy a textbook, as we will be using quite a few different sources of information. I will make excerpts available for class. Here is a short list of references we will be using, in case you want a sneak preview:

- Hilty, Lorenz M., and Bernard Aebischer. ICT Innovations for Sustainability. Springer International, 2015.
- Booth Sweeney, L., Meadows, D. The systems thinking playbook: Exercises to Stretch and Build Learning and Systems Thinking Capabilities. Chelsea Green Publishing; 2010.
- Purdue University Online Writing Lab. 2015. Purdue University. (https://owl.english.purdue.edu)

Class notes, assigned readings, assignment descriptions, rubrics, and further resources will be available on Beachboard. Please let me know in case you are missing anything.

COURSE ETIQUETTE

All discussion is welcome and encouraged in class as long as it is respectful. We all continue to learn and grow every day. Feedback goes both ways, so I strongly welcome feedback about what works well in class and where you struggle. I will encourage you to set and track your own learning goals, and I will offer several modalities for how to do certain assignments and how to show your contributions to class because not everyone learns the same way. I want you to be able to show your efforts for this course in an outlet medium that suits you. This means that some of the assignments for the required writing can have different forms depending on what styles you are comfortable with and want to practice, as long as you fulfill the baseline requirements.
I encourage you to ask in class first for questions that may be relevant for other students as well, but you can always write me an email as well. Please come see me during office hours or make an appointment with me to discuss individual challenges that need more time, or if your concern is about your specific solution for an assignment.

COURSE REQUIREMENTS

All assignments are designed to help you deepen you understanding of the content discussed in the lectures and to help you develop the skills you need to engage with the different analysis aspects of sustainability. I allow for quite a bit of teamwork in my courses, but you have to always make known who contributed what to a team assignment. You can ask me to give a presentation in class as well; there will be a couple of opportunities for that.

WRITTEN TASKS

- **Freewriting**: You will complete 5 freewriting tasks (approximately 1 page each), to begin exploring some key concepts and projects for this class. This gives you a lightweight introduction into writing and into putting your own reflective thoughts into words.
  1. Sustainability over the course of history
  2. Sustainability in relation to IT
  3. Exploring an example sustainability project

4. Impacts of human systems on the planet
5. Vision of a sustainable society

- **Definition**: Write your own definition of sustainability (200 words). This will help you reflect on the different definitions that you have heard and to compare and decide in which direction your personal understanding of the term develops.

- **Individual papers**: You will write 3 individual papers over the course of the semester, two with 1500 words and one with 1800 words. They let you explore concepts in more depth and work on your academic writing style.
  1. 5-Dimension Analysis: Analyze the five dimensions of sustainability for a case study (1500w)
  2. Leverage points analysis: Analyze leverage points for an example system (1500w)
  3. Final essay on chosen special topic with instructor consent (1800w)

PRESENTATION / DISCUSSION TASKS

- **Scenario exploration**: Discussion of scenarios with cultural factors, establish sustainability policies amongst a team and sum them up according to a template for the other teams at the end of the activity.
- **Presentation**: Oral presentation of 10 minutes duration on a selected topic (instructor will provide list to choose from) in class with a 1-page handout summarizing the key points.
- **Active participation**: You are expected to actively participate in class and in team discussions – it makes you learn better, reflect more, and trains your communication skills.

GRADING

Assignment	***Output***	***Revision***	***Final version***
Definition of sustainability (individual)	200 words	5%	5%
5-Dimension analysis (individual)	1500 words	5%	5%
Scenario exploration (team)	Oral summary	n.a.	10%
Oral presentation (team)	Slides & Handout	n.a.	20%
Leverage points analysis (individual)	1500 words	10%	10%
Final essay (individual)	1800 words	10%	20%

Base calculation for grades: 90-100: A, 80-89: B, 70-79: C, 60-59: D, <60 F.
You can see that the points are fairly well distributed throughout the semester with a slight emphasis on your final essay. This means even if the first couple of assignments don't go so well, you can still get an A in this course. Furthermore, you can score a quite a bit with active contribution in class discussions and activities.

Attendance policies and provision for makeup of assignments: I expect continuous course attendance outside of excused absence. No late work will be accepted outside of excused exceptions that were given at least 2 days before the deadline. The benefit of attending class regularly is that you can use most of the activity time to work towards your assignments and that you have input from all the discussions in class.

UNIVERSITY POLICIES

Academic Honesty:
Cheating and plagiarism will not be tolerated in this course. Any individual caught cheating on quizzes, homework, lab projects, or the final exam will be punished to the full extent allowed under University regulations. Plagiarism on papers or assignments is not acceptable and work that is

plagiarized will not receive credit. Plagiarism is considered cheating. Note: any time another person's work is used without giving them proper credit, it is considered plagiarism and cheating. It is also considered plagiarism if you try to reuse work from other courses for the deliverables in this course. At a minimum, any student caught cheating will receive no credit for the work concerned, and will receive a reduction of one letter grade from their final course grade. The official CSULB Policy on Cheating and Plagiarism can be found here:
http://web.csulb.edu/divisions/aa/catalog/current/academic_information/cheating_plagiarism.html

Reasonable Accommodation:

Individuals with disabilities who need assistance or modification to the University's programs and/or activities should inform the person(s) responsible for these programs and/or activities immediately upon knowing that such modification is necessary. Individuals registered with the California Department of Rehabilitation may be eligible for assistance through that agency. Students may be eligible for assistance through the Office of Disabled Student Services, located in Brotman Hall 270, telephone (562) 985-5401. For evaluation and service, contact that office directly. If the modification or accommodation provided is inappropriate or insufficient, you may seek the assistance of the Office of Equity and Diversity, located in University Student Union 301, telephone (562) 985-8256. If a reasonable accommodation has been requested but was not provided, the individual may access the complaint resolution process.

Basic Needs:

Any student who has difficulty affording groceries or accessing sufficient food to eat every day, or who lacks a safe and stable place to live, and believes this may affect their performance in the course, is urged to contact the Dean of Students for support. Furthermore, please notify the professor if you are comfortable in doing so. This will enable her to provide any resources that she may possess.

COE Tutoring Center Announcement

Take advantage of free peer tutoring to keep up your grades in the most challenging classes. Tutoring is available for undergraduate engineering students in departmental courses for Electrical Engineering, Mechanical & Aerospace Engineering, Civil Engineering, Computer Engineering/Science, and Chemical Engineering.
Tutoring is on a first-come-first-serve, walk-in basis. Tutors are available Monday-Thursday in the Fall and Spring terms. All tutoring sessions take place in the Engineering Student Success Center (ESSC) in EN2, Room 300 between the hours of 9:00 a.m. – 6:00 p.m.
Visit the website for detailed tutoring schedules:
http://web.csulb.edu/colleges/coe/views/essc/academic_success/engineering_tutor.shtml#asp_ETP

DETAILED TIMELINE (TENTATIVE, SOME DETAILS MAY CHANGE AS WE PROCEED)

Week	Lecture a	Activity a	Deliverable a	Lecture b	Activity b	Deliverable b
1: The concept of sustainability	Introduction: Sustainability as it developed over time in history	Introduction to freewriting, draft of 1st freewriting task (1 page)	Freewrite 1 due	The most common definitions and how they relate	Brainstorm on definitions in teams, submit draft of 200 words	Draft definition due
2: ICT and sustainability	Sustainability in ICT and ICT for sustainability	Find more examples and classify them, 2nd freewriting task (1 page)	2nd freewrite due, plus revision of freewrite 1	Renewable energy and smart grids: production and potentials	Discussion on renewable energy production in CA	Revision of Sustainability Definition due
3: The five dimensions of sustainability	The five dimensions of sustainability	Analyze the five dimensions for a case study, discussion with peers	Revision of freewrite 2	Student mini presentations on case study	Rough draft workshop on 5-Dimension Analysis paper, editing to be completed as homework	5-Dimension Analysis 1000 word draft, feedback by Activity 4b, revision due by Activity 5b
4: The orders of effect and rebound effects	The orders of effect (LES model)	Analyze the orders of effect for a case study, discuss in team	-	Rebound effects	Analyze potential rebound effects in case study, discuss in team	
5: Peak oil and climate change	Peak Oil presentation	Discussion on peak oil, renewable energy, future alternatives		Climate change	Discuss "An inconvenient truth", reactions and follow-up initiatives	Revision of 5-Dimension Analysis due
6: Sustainable food production and permaculture	Sustainable agriculture *(available for student presentations)*	Team discussion on food habits	Handout *(for those students who did a presentation)*	Permaculture *(available for student presentations)*	Discussion: permaculture and relation to our high-tech society	Handout *(for those students who did a presentation)*
7: Smart X – green product and service development	Overview of smart technology and service development	Individual research on gaps and open challenges in smart technology	Topic idea for final essay	Smart systems *(available for student presentations)*	Reflection on effectiveness of presentation and discussion on final essay ideas	Handout *(for those students who did a presentation)*
8: Behavior change and gamification of sustainability	Behavior change	3rd freewrite task	Freewrite 3 due,	Gamification	Freewrite 3 feedback; Discuss Jane McGonagall TED talk	-
9: Sustainable development concepts	Concepts - The Karlskrona Manifesto	Comparing different designs / solutions, freewrite task 4	Revision of freewrite 3, draft of freewrite 4	Concepts of decoupling, dematerialization and substitution	Find other solutions and discuss them, short student presentations	-
10: Sustainable development process	Sustainable development with the IMAGINE approach	Case study analysis: Try out with scenario, short student presentations	Revision of freewrite 4	Organizational and cultural factors during development	Group writing: Play through a number of scenarios with cultural factors, establish policies	-
11. Introduction to systems	Introduction to Systems thinking	Exercises from systems thinking playbook	-	Recap of the Sustainable Development	Free writing for topic ideas for the final essay,	-

thinking				Goals	followed by peer feedback session	
12. System dynamics and leverage points	Introduction to System dynamics	Case study analysis: System dynamics modeling of example case study, freewrite task 5	Draft of freewrite 5	System dynamics continued	Rough draft workshop for the final essays with feedback by peers	-
13: Leverage points	Leverage points overview	Case study analysis: Analyze leverage points for sample system, start drafting Leverage Points paper		Leverage points interaction and in-depth	Group writing: Rework and refine leverage points for sample system	Complete 1500 word draft of Leverage Points paper by end of week
14: Sustainability Assessment	Measuring & Metrics and ISO standards (ISO 14000 & 26000)	Define metrics in all dimensions for a case study and discuss in team	-	Introduction of LCA	Try out LCA with a small example, short student presentations	-
15: Future concepts, saving the world, and the like	Utopias	Develop your own scenario, short student presentations	Revision Leverage Points paper due	Latest trends in research and innovations in sustainability	Write your own research questions and discuss research design with peers	Final essay due by end of the week

If you made it all the way down to here – congrats – please email me, introduce yourself and send me a picture of your winter break ☺